

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I Atmakuri Sai Mrudula (192372087) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Mrudula

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly	Uses relevant	Uses vague or	No useful

		relevant cloud examples to strengthen the answer	examples with minor mismatch	partially relevant examples	examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud Computing: Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, and artificial intelligence over the Internet. Instead of purchasing and maintaining physical hardware, users can access these resources whenever needed from cloud service providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform. Cloud computing follows a pay-as-you-go model, allowing users to pay only for the resources they use. This approach reduces infrastructure costs, increases flexibility, and improves business efficiency.

Traditional on-premise computing requires organizations to purchase, install, and maintain their own servers, storage devices, networking equipment, and software within their premises. It involves high capital investment, regular maintenance, and dedicated IT staff. Cloud computing eliminates these responsibilities by allowing cloud providers to manage the infrastructure while users focus on their applications and business operations.

Characteristics

1. On-Demand Self-Service

- Users can provision computing resources whenever required without human interaction from the service provider.
- Example: Creating a virtual machine in minutes on AWS.

2. Broad Network Access

- Services are accessible from anywhere using devices such as laptops, smartphones, and tablets through the Internet.
- Enables remote work and collaboration.

3. Resource Pooling

- Cloud providers share computing resources among multiple customers using a multi-tenant architecture.
- Resources are dynamically allocated based on demand.

4. Rapid Elasticity

- Resources can be scaled up or down automatically depending on workload.
- Example: An e-commerce website increases server capacity during festival sales.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

The evolution of cloud computing is the result of continuous advancements in computer hardware, networking, virtualization, and Internet technologies. Initially, organizations relied on mainframe computers, where a single powerful system served multiple users through connected terminals. Although mainframes provided centralized computing, they were expensive and difficult to maintain.

To improve hardware utilization, server virtualization was introduced. Virtualization allowed multiple virtual machines to run on a single physical server, reducing hardware costs and improving resource efficiency. Later, distributed computing, including cluster computing and grid computing, enabled multiple computers to work together to solve complex computational problems and share processing power.

Evolution Timeline

1. Hardware Evolution (1950s–1970s)

- Large mainframe computers served multiple users.
- Time-sharing allowed many users to share one system.

2. Personal Computing Era (1980s)

- Personal computers became common.
- Computing shifted from centralized systems to individual devices.

3. Networking and Client-Server Computing (1990s)

- LAN and Internet technologies connected computers.
- Client-server architecture became popular.

4. Virtualization Technology (Late 1990s)

- Hypervisors enabled multiple virtual machines on one physical server.
- Improved hardware utilization and reduced costs.

5. Internet Software Evolution (2000s)

- Web applications and Software as a Service (SaaS) emerged.
- High-speed Internet enabled online services.

6. Modern Cloud Computing (Today)

- Providers like AWS, Microsoft Azure, and Google Cloud offer scalable infrastructure and services.
- Supports AI, IoT, Big Data, DevOps, and serverless computing.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization is a technology that allows a single physical server to be divided into multiple virtual machines (VMs), each running its own operating system and applications. It improves hardware utilization, reduces infrastructure costs, simplifies server management, and enables efficient resource allocation. Virtualization is the foundation of cloud computing because it allows cloud providers to offer scalable and flexible computing resources.

This is critical to cloud computing because it enables resource pooling many tenants sharing one physical server securely, elastic provisioning VMs created/resized/destroyed in minutes, better hardware utilization (avoiding idle servers), and fault isolation a crash in one VM doesn't affect others.

Role of Server Virtualization

Server virtualization divides one physical server into multiple Virtual Machines (VMs) using a hypervisor.

Each VM:

- Runs its own operating system.
- Has dedicated CPU, memory, and storage.
- Operates independently.

Benefits

- Better hardware utilization
- Lower infrastructure cost
- Faster deployment
- Easy backup and recovery
- High availability
- Simplified resource management

Differentiation:

Virtualization is the enabling technology it operates at the infrastructure level, abstracting hardware into virtual resources e.g., running 5 VMs on one physical server. Cloud computing is the service model built on top of that technology it delivers those virtualized resources to customers on-demand, over the internet, with metered billing e.g., renting a VM instance from AWS EC2. In short: virtualization is the "how," cloud computing is the "what" is delivered as a service. Virtualization can exist without cloud computing e.g., a company virtualizing its own servers on-premise, but public cloud computing would not be commercially viable without it.

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

SOAP (Simple Object Access Protocol)

SOAP (Simple Object Access Protocol) is a strict, protocol-based messaging standard. It uses XML only, wrapped in a formal envelope structure (Header + Body), and requires a WSDL contract defining available operations. It supports stateful operations and has strong built-in security (WS-Security), making it suitable for enterprise-grade transactions, but it's verbose and slower due to XML overhead.

REST (Representational State Transfer)

REST (Representational State Transfer) is an architectural style that uses standard HTTP methods like GET, POST, PUT, DELETE to act on resources identified by URLs. It supports flexible data formats commonly JSON, but also XML or plain text and is stateless, meaning each request carries all necessary information with no server-side session. It relies on transport-level security HTTPS, tokens/API keys and is lightweight, fast, and easy to integrate with web/mobile apps.

Suitability for cloud applications:

REST is the dominant choice for modern cloud services (AWS, Azure, Google Cloud APIs) because its statelessness fits well with elastic, horizontally scalable cloud architectures, and it's lightweight and easy to use. SOAP remains preferred in enterprise/legacy systems requiring formal contracts and high security compliance, such as banking, healthcare, and government applications.

Communication Style:

- **SOAP:** A protocol that follows strict communication standards.
- **REST:** An architectural style that uses HTTP methods (GET, POST, PUT, DELETE).

Data Exchange:

- **SOAP:** Uses only XML format.
- **REST:** Supports JSON, XML, HTML, and plain text, with JSON being the most common.

Performance:

- **SOAP:** Slower due to XML processing.
- **REST:** Faster and lightweight.

Suitability for Cloud Applications:

- **SOAP:** Best suited for secure enterprise applications such as banking and healthcare.
- **REST:** Best suited for cloud, web, and mobile applications because it is simple, scalable, and efficient.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud computing provides different service models based on the level of control and management offered to users.

1. Infrastructure as a Service (IaaS)

- IaaS provides virtualized computing resources such as servers, storage, and networking over the Internet.
- Users are responsible for managing the operating system, applications, and data, while the cloud provider manages the physical infrastructure.
- **Example:** AWS EC2.

2. Platform as a Service (PaaS)

- PaaS provides a complete platform for developing, testing, and deploying applications.
- The cloud provider manages the infrastructure, operating system, and runtime environment, allowing developers to focus only on coding.
- **Example:** Google App Engine.

3. Software as a Service (SaaS)

- SaaS delivers ready-to-use software applications through the Internet.
- Users can access the software using a web browser without installing or maintaining it.
- The service provider manages the application, updates, and security.
- **Example:** Microsoft 365.

4. Anything as a Service (XaaS)

- XaaS (Everything/Anything as a Service) is a broad cloud model that provides various IT services such as databases, security, storage, networking, and artificial intelligence over the Internet.
- It allows organizations to use different services on a subscription or pay-as-you-use basis.
- **Example:** Database as a Service (DBaaS).